



3.1 The Chain Rule

WARM UP!

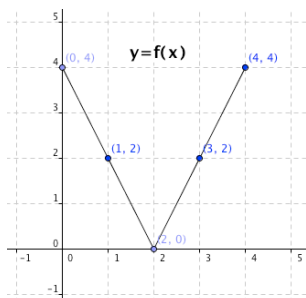
Review of Function Compositions

$$(f \circ g)(x) = f(g(x))$$

$$(g \circ f)(x) = g(f(x))$$

Outer function
 $y = f(g(x))$
Inner function

Ex 1: Evaluate.



$$g(x) = x - 5$$

x	$h(x)$
-3	1
2	6
4	0
5	2

a) $f(g(6))$ 2	b) $h(f(4))$ 0	c) $f(h(5))$ 0	d) $f(h(2))$ 2
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Ex 2: Identify the outside function, $f(u)$, and inside function, $g(x)$.

a) $y = (3x - 5)^2$ $f(u) = u^2$ $g(x) = 3x - 5$	b) $y = \sin 2x$ $f(u) = \sin u$ $g(x) = 2x$	c) $y = \sqrt{x + 3}$ $f(u) = \sqrt{u}$ $g(x) = x + 3$	d) $y = \sin^2 x$ $f(u) = u^2$ $g(x) = \sin x$
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The Chain Rule

Use the chain rule to differentiate composite functions.

Ex 3: Circle functions that require the use of chain rule to differentiate.

$$y = 7x^3 + 1$$

$$y = x^2 + 1$$

$$y = \sin 2x$$

$$y = e^{-x}$$

$$y = 3x - 1$$

$$y = x \tan x$$

$$y = \sqrt{x^2 + 1}$$

$$y = (7x + 1)^3$$



***Teaching Tip:** Give students layered candy, such as peanut M&Ms (be mindful of peanut allergies). Tell the students to place one M&M in their mouth. Tell the students NOT to chew the M&M yet, just to observe what happens. Simultaneously, teach the chain rule. Then relate the rule with the M&M. While the candy layer melts, the chocolate and peanut stay the same. Then the chocolate melts but the peanut stays intact. Last, you must take care of the peanut (i.e. chew the peanut) or bad things will happen. **Don't choke!** As you teach the example 4, refer back to the peanut M&M.





Ex 4: Find the derivative.

a) $y = (x^2 + 1)^3$ $y' = 6x(x^2 + 1)^2$	b) $y = \frac{-7}{(2x-3)^2}$ $y' = \frac{28}{(2x-3)^3}$
c) $y = \sin 2x$ $y' = 2 \cos 2x$	d) $y = \tan^2 x$ $y' = 2 \tan x \sec^2 x$
e) $y = \ln(2x)$ $y' = \frac{1}{x}$	f) $y = (\ln x)^3$ $y' = \frac{3(\ln x)^2}{x}$
g) $y = e^{2x^3-3x}$ $y' = (6x^2 - 3)e^{2x^3-3x}$	h) $y = \frac{1}{e^x}$ $y' = -\frac{1}{e^x}$

Ex 5:

If $f(x) = \sin(\ln(2x))$, then $f'(x) =$

(A) $\frac{\sin(\ln(2x))}{2x}$

(B) $\frac{\cos(\ln(2x))}{x}$

(C) $\frac{\cos(\ln(2x))}{2x}$

(D) $\cos\left(\frac{1}{2x}\right)$



Ex 6:

If $f(x) = e^{2x}(x^3 + 1)$, then $f'(2) =$

- (A) $6e^4$ (B) $21e^4$ (C) $24e^4$ (D) $30e^4$

Ex 7: 2023 AB 5 – part (a)

x	0	2	4	7
$f(x)$	10	7	4	5
$f'(x)$	$\frac{3}{2}$	-8	3	6
$g(x)$	1	2	-3	0
$g'(x)$	5	4	2	8

5. The functions f and g are twice differentiable. The table shown gives values of the functions and their first derivatives at selected values of x .

a) Let h be the function defined by $h(x) = f(g(x))$. Find $h'(7)$. Show the work that leads to your answer.

$$h'(x) = f'(g(x)) \cdot g'(x)$$

Chain rule

1 point

$$h'(7) = f'(g(7)) \cdot g'(7)$$

$$= f'(0) \cdot 8 = \frac{3}{2} \cdot 8 = 12$$

Answer

1 point

2023 Scoring Guidelines

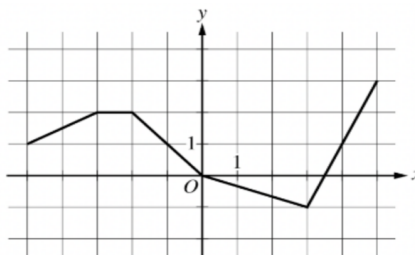
SCORING STATS!

Guess the mean: 4.46/9

2023 Scoring Statistics

Ex 8: 2017 AB6 – parts (b) & (c)

x	$g(x)$	$g'(x)$
-5	10	-3
-4	5	-1
-3	2	4
-2	3	1
-1	1	-2
0	0	-3



Graph of h

SCORING STATS!

Guess the mean: 3.32/9

2017 Scoring Statistics

Let f be the function defined by $f(x) = \cos(2x) + e^{\sin x}$.

Let g be a differentiable function. The table above gives values of g and its derivative g' at selected values of x .

Let h be the function whose graph, consisting of five line segments, is shown in the figure above.

(b) Let k be the function defined by $k(x) = h(f(x))$. Find $k'(\pi)$.

(c) Let m be the function defined by $m(x) = g(-2x) \cdot h(x)$. Find $m'(2)$.

$$(b) k'(x) = h'(f(x)) \cdot f'(x)$$

$$\begin{aligned} k'(\pi) &= h'(f(\pi)) \cdot f'(\pi) = h'(2) \cdot (-1) \\ &= \left(-\frac{1}{3}\right)(-1) = \frac{1}{3} \end{aligned}$$

$$2: \begin{cases} 1: k'(x) \\ 1: k'(\pi) \end{cases}$$

$$(c) m'(x) = -2g'(-2x) \cdot h(x) + g(-2x) \cdot h'(x)$$

$$\begin{aligned} m'(2) &= -2g'(-4) \cdot h(2) + g(-4) \cdot h'(2) \\ &= -2(-1)\left(-\frac{2}{3}\right) + 5\left(-\frac{1}{3}\right) = -3 \end{aligned}$$

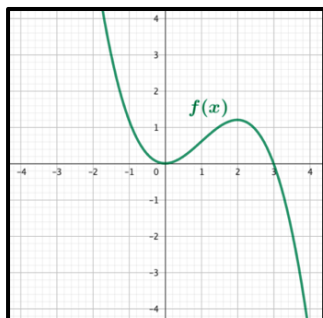
$$3: \begin{cases} 2: m'(x) \\ 1: m'(2) \end{cases}$$

2017 Scoring Guidelines



Ex 9: *Medium*

Let $y = f(x^2)$. Use the graph of f given below to determine the sign of $\frac{dy}{dx}\bigg|_{x=-1}$.



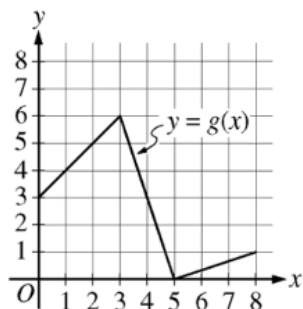
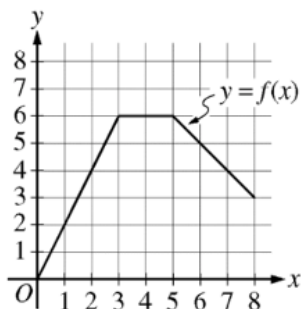
$$\frac{dy}{dx} = f'(x^2) \cdot 2x$$

$$\frac{dy}{dx}\bigg|_{x=-1} = f'((-1)^2) \cdot 2(-1)$$

$$\frac{dy}{dx}\bigg|_{x=-1} = f'(1) \cdot -2$$

$$\frac{dy}{dx}\bigg|_{x=-1} = + \cdot - < 0$$

Ex 10: Use the graphs of f and g below to evaluate each expression.



a) *Spicy* 🌶️🌶️

$$\lim_{x \rightarrow 2} \frac{g(3x) - g(6)}{x - 2}$$

$$\text{Alternate Form: } f'(a) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$$

Let $y = g(3x)$ and $x = 2$.

$$\frac{dy}{dx} = g'(3x) \cdot 3$$

$$\frac{dy}{dx}\bigg|_{x=2} = g'(3 \cdot 2) \cdot 3$$

$$\frac{dy}{dx}\bigg|_{x=2} = g'(6) \cdot 3$$

$$\frac{dy}{dx}\bigg|_{x=2} = \frac{1}{3} \cdot 3 = 1$$

b) *Spicy* 🌶️🌶️

$$\lim_{h \rightarrow 0} \frac{f(2(1+h)) - f(2)}{h}$$

Limit Definition of the Derivative:

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

Let $y = f(2x)$ and $x = 1$.

$$\frac{dy}{dx} = f'(2x) \cdot 2$$

$$\frac{dy}{dx}\bigg|_{x=1} = f'(2 \cdot 1) \cdot 2$$

$$\frac{dy}{dx}\bigg|_{x=1} = f'(2) \cdot 2$$

$$\frac{dy}{dx}\bigg|_{x=1} = 2 \cdot 2 = 4$$